

# BAUST - Department of Computer Science & Engineering

## CSE 2103: Database Systems | Semester Final Exam - Fall 2023 | Batch 11

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**Instructions: Answer any 4 questions out of 6. Figures indicate full marks.**

### Question 1: ER Modeling and Relational Mapping [17.5 Marks]

- a) Design an ER diagram for a Hospital Management System containing Patients, Doctors, Appointments, and Prescriptions. Identify key constraints.
- b) Map the ER model into normalized relational schemas (1NF/2NF/3NF).

### Question 2: Relational Algebra & Advanced SQL [17.5 Marks]

Consider the schema: Student(id, name, dept), Enrolls(id, courseCode, grade).

- a) Write a relational algebra expression to find names of students enrolled in all CSE courses.
- b) Write an optimized SQL query using window functions (DENSE\_RANK) to find top 3 students.

### Question 3: Functional Dependencies and Normalization [17.5 Marks]

Given relation R(A, B, C, D, E) with FDs: { A → BC, CD → E, B → D, E → A }

- a) Find all candidate keys of R.
- b) Decompose R into BCNF. State whether decomposition is dependency preserving.

### Question 4: Transaction Processing & Concurrency [17.5 Marks]

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- a) Explain ACID properties with real-world banking failure scenarios.
- b) Differentiate between Two-Phase Locking (2PL) and Strict 2PL regarding cascading rollbacks.